# Condensed Matter Notes

# 1 Lattice vibrations

$$\boldsymbol{r}_{n\alpha}(t) = \boldsymbol{R}_{n\alpha}^{(0)} + \boldsymbol{u}_{n\alpha}(t) \tag{1.1}$$

$|\boldsymbol{u}| \ll a$ where $a$ is lattice constant.

$$L = T - U = \frac{1}{2} \sum_{n\alpha} M_\alpha \left( |\boldsymbol{u}|_{n\alpha}^2 - U(\{\boldsymbol{u}\}) \right) \tag{1.2}$$

$$U = U_0 + \frac{1}{2} \sum_{n,\alpha,i;m,\beta,j} \left. \frac{\partial^2 U}{\partial u_{n\alpha,i} \partial u_{m\beta,j}} \right|_0 u_{n\alpha,i} u_{m\beta,j} \tag{1.3}$$

Where $i, j \in \{1, 2, ..., d\}$ is the index of the spatial dimension.

$$\Phi_{n\alpha,i;m\beta,j} \equiv \left. \frac{\partial^2 U}{\partial u_{n\alpha,i} \partial u_{m\beta,j}} \right|_0 \tag{1.4}$$

$$M_\alpha \ddot{u} = -\sum_{m,\beta,j} \Phi_{n\alpha,i;m\beta,j} u_{m\beta,j} \tag{1.5}$$

# 2 Coherent states

## 2.1 Bosonic coherent states

We begin by defining the displacement operator that generates the coherent state from the vacuum:

$$D\left(\hat{\phi}\right) \equiv \exp\left( \sum_\beta \phi_\beta \hat{a}_\beta^\dagger \right) \tag{2.6}$$

Our goal is to show that this state satisfies the eigenvalue equation $\hat{a}_\alpha |\phi\rangle = \phi_\alpha |\phi\rangle$.

To proceed, we need a useful identity for commutators involving exponentials. When two operators $\hat{A}$ and $\hat{B}$ have a commutator $\left[\hat{A}, \hat{B}\right] = c$ that is simply a c-number (a complex number rather than an operator), we can establish that:

$$\left[\hat{A}, e^{\hat{B}}\right] = \left[\hat{A}, \hat{B}\right] e^{\hat{B}} \tag{2.7}$$

This follows from expanding the exponential as $e^{\hat{B}} = \sum_{n=0}^{\infty} \hat{B}^n / n!$ and using the property $\left[\hat{A}, \hat{B}^n\right] = n\left[\hat{A}, \hat{B}\right]\hat{B}^{n-1}$, which holds when the commutator is a c-number.

Now we apply this identity to our specific case. Taking $\hat{A} = \hat{a}_\alpha$ and $\hat{B} = \sum_\beta \phi_\beta \hat{a}_\beta^\dagger$, we first compute their commutator:

$$\left[\hat{a}_\alpha, \sum_\beta \phi_\beta \hat{a}_\beta^\dagger\right] = \sum_\beta \phi_\beta \left[\hat{a}_\alpha, \hat{a}_\beta^\dagger\right] = \sum_\beta \phi_\beta \delta_{\alpha\beta} = \phi_\alpha \tag{2.8}$$

$$\langle\phi|\phi'\rangle = \left\langle 0 \middle| \exp\left(\sum_\alpha \phi_\alpha^* \hat{a}_\alpha\right) \middle| \phi' \right\rangle = \exp\left(\sum_\alpha \phi_\alpha^* \phi_\alpha\right) \langle 0|\phi\rangle = \exp(\phi^* \cdot \phi) \tag{2.9}$$

$$\hat{\alpha}^\dagger |\phi\rangle = \hat{\alpha}^\dagger \exp\left(\sum_\alpha \phi_\alpha \hat{a}_\alpha^\dagger\right) |0\rangle = \frac{\partial}{\partial \phi_\alpha} \exp\left(\sum_\alpha \phi_\alpha \hat{a}_\alpha^\dagger\right) |0\rangle = \frac{\partial}{\partial \phi_\alpha} |\phi\rangle \tag{2.10}$$

$$\left[\hat{a}_\alpha^\dagger, |\phi\rangle\langle\phi|\right] = \left(\frac{\partial}{\partial \phi_\alpha} - \phi_\alpha^*\right) |\phi\rangle\langle\phi| \tag{2.11}$$

The standard Gaussian measure on the complex plane:

$$d^2\phi = d(\operatorname{Re}\phi)(\operatorname{Im}\phi) \Rightarrow \int e^{-|\phi|^2} d^2\phi = \pi \tag{2.12}$$

Sometimes, we prefer to use $d\phi d\phi^* = 2i d^2\phi$:

$$d\mu(\phi) = \prod_\alpha \frac{\mathrm{d}\phi_\alpha \, \mathrm{d}\phi_\alpha^*}{2\pi i} \exp\left(-\sum_\alpha |\phi_\alpha|^2\right) \tag{2.13}$$

$$\int d\mu(\phi) \left[a_\alpha^\dagger, |\phi\rangle\langle\phi|\right] = \int d\mu(\phi) \left(\frac{\partial}{\partial \phi_\alpha} - \phi_\alpha^*\right) |\phi\rangle\langle\phi| \tag{2.14}$$

Use integral by parts:

$$\frac{\partial}{\partial \phi_\alpha} \exp\left(-\sum_\alpha |\phi_\alpha|^2\right) = -\phi_\alpha^* \exp\left(-\sum_\alpha |\phi_\alpha|^2\right) \tag{2.15}$$

So everything cancelled. Since the measure commutes with every $a_\alpha, a_\alpha^\dagger$. By full Fock space irreducibility, Schur' s lemma tells us that the measure must be proportional to the identity $O = c\mathbf{1}$.

$$
\begin{aligned}
&\hat{a}_\alpha|\phi\rangle = \phi_\alpha|\phi\rangle = \frac{\partial}{\partial\phi^*}|\phi\rangle \\
&\langle\phi|\hat{a}_\alpha^\dagger = \langle\phi|\phi_\alpha^*
\end{aligned}
\tag{2.16}
$$

$$
|\psi\rangle = \int d\mu(\phi)\psi(\phi^*)|\phi\rangle \tag{2.17}
$$

$$
\hat{a}_\alpha \to \frac{\partial}{\partial\phi^*}, \quad \hat{a}_\alpha^\dagger \to \phi_\alpha^* \tag{2.18}
$$

It' s clear that the unitary transformation:

$$
\langle\phi'|\psi\rangle = \int d\mu(\phi)\langle\phi'|\phi\rangle\langle\phi|\psi\rangle = \int d\mu(\phi)\exp(\phi'\cdot\phi)\psi(\phi) \tag{2.19}
$$

$$
\left\langle\phi'\middle|\hat{A}(\hat{a}^\dagger,\hat{a})\middle|\phi\right\rangle = A(\phi^*,\phi)\langle\phi'|\phi\rangle = A(\phi^*,\phi)e^{\phi'\cdot\phi} \tag{2.20}
$$

## 2.2 Grassmann variables

Let $\xi, \xi^*$ be independent Grassmann variables.

$$
[\xi,\xi^*]_+ = 0, \quad \xi^2 = 0, \quad (\xi^*)^2 = 0 \tag{2.21}
$$

$$
e^\xi = \sum_{i\geq 0}\frac{1}{i!}\xi^i = 1 + \xi \Rightarrow f(\xi) = f_0 + f_1\xi \tag{2.22}
$$

Because the translation invariance, the total integral at boundary is zero:

$$
\begin{aligned}
&\int d\xi f(\xi+\eta) = \int d\xi f(\xi) \\
&aI_1 + bI_\xi + b\eta I_1 = aI_1 + bI_\xi \Rightarrow I_1 = 0
\end{aligned}
\tag{2.23}
$$

$$
I_\xi = B \Rightarrow \text{choose } B = 1 \tag{2.24}
$$

$$
\frac{\partial}{\partial\xi}(\xi^*\xi) = \frac{\partial}{\partial\xi}(-\xi\xi^*) = -\xi^* \tag{2.25}
$$

$$
\begin{aligned}
\int d\xi^* d\xi e^{-\xi^*\xi} f^*(\xi) g(\xi^*) &= \int d\xi^* d\xi (1-\xi^*\xi)(f_0^* + f_1^*\xi)(g_0 + g_1\xi^*) \\
&= f_0^* g_0 + f_1^* g_1
\end{aligned}
\tag{2.26}
$$

It' s worth noting:

$$\xi_\alpha|0\rangle = \xi_\alpha(1-\xi_\alpha)|0\rangle = \xi_\alpha\left(1-\xi_\alpha\hat{a}_\alpha^\dagger\right)|0\rangle \tag{2.27}$$

$$|\xi\rangle = \exp\left(-\sum_\alpha \xi_\alpha\hat{a}_\alpha^\dagger\right)|0\rangle = \prod_\alpha\left(1-\xi_\alpha\hat{a}_\alpha^\dagger\right)|0\rangle \tag{2.28}$$

$$\hat{a}_\alpha|\xi\rangle \sim \hat{a}_\alpha - \hat{a}_\alpha\xi_\alpha\hat{a}_\alpha^\dagger|0\rangle = \xi_\alpha|0\rangle = \xi_\alpha\left(1-\xi_\alpha\hat{a}_\alpha^\dagger\right)|0\rangle \Rightarrow \hat{a}_\alpha|\xi\rangle = \xi_\alpha|\xi\rangle \tag{2.29}$$

# 3 Finite-temperature field theory

## 3.1 Thermal evolution

For general bracket:

$$\left[\hat{a}_\alpha, \hat{a}_\beta^\dagger\right]_\mp = \delta_{\alpha\beta} \tag{3.30}$$

$$\hat{\rho} = \exp\left(-\beta\left(\hat{H}-\mu\hat{N}\right)\right)/Z \tag{3.31}$$

$$\begin{aligned}\left\langle\hat{O}(t)\right\rangle = \operatorname{tr}\left[\hat{\rho}\,\hat{O}(t)\right] &= \operatorname{tr}\left[\hat{\rho}e^{-i\hat{H}t}\hat{O}e^{i\hat{H}t}\right] \\ &= \operatorname{tr}\left[e^{i\hat{H}t}\hat{\rho}e^{-i\hat{H}t}\hat{O}\right] \\ &= \operatorname{tr}\left[\hat{\rho}\hat{O}\right] = \left\langle\hat{O}(0)\right\rangle\end{aligned} \tag{3.32}$$

For $\hat{H}$ independent of time. The last step is due to $\hat{\rho}$ commutes with $e^{i\hat{H}t}$.

$$G^{(n)}(\alpha_i t_i, \alpha_i' t_i') = (-i)^n\left\langle\hat{T}\left[\prod_{i=1}^{n} a_{\alpha_i}^{(H)}(t_i)\prod_{i=n}^{1} a_{\alpha_i'}^{\dagger(H)}(t_i')\right]\right\rangle \tag{3.33}$$

- $a_\alpha^{(H)}(t)$ is annihilation operator in Heisenberg picture.
- $\langle...\rangle$ is ensemble or ground state average.
- $\hat{T}(...)$ is time ordering operator: $\hat{T}\left[\prod_i \hat{O}(t_i)\right] = \prod_i \hat{O}\left(t_{\rho(i)}\right)$ where $\rho(i)$ is any possible time permutation that make it ascending or descending based on the context and goal.

$$G(\alpha t; \alpha' t) = -i\left\langle\hat{T}\left[\hat{a}_{\alpha_1}(t)\hat{a}_{\alpha_1'}^\dagger(t')\right]\right\rangle \tag{3.34}$$

$$\hat{a}_\alpha^{(H)}(t) = e^{iHt}\hat{a}_\alpha e^{-iHt} \tag{3.35}$$

$$\begin{aligned}\left\langle \hat{a}_\alpha^{(H)}(t)\hat{a}_\alpha^{\dagger(H)}(t')\right\rangle &= \sum_n \rho_n \left\langle n\middle|\hat{a}_\alpha^{(H)}(t)\hat{a}_\alpha^{\dagger(H)}(t')\middle|n\right\rangle \\ &= \sum_n \rho_n \left\langle n\middle|e^{iHt}\hat{a}_\alpha e^{-iH(t-t')}\hat{a}_\alpha^\dagger e^{-iHt}\middle|n\right\rangle\end{aligned} \tag{3.36}$$

Suppose: $|\psi_n(t)\rangle = e^{-iHt}|n\rangle$:

$$\text{LHS} = \sum_n \rho_n \left\langle \psi_n(t)\middle|\hat{a}_\alpha e^{-iH(t-t')}\hat{a}_\alpha^\dagger\middle|\psi_n(t')\right\rangle \tag{3.37}$$

Assuming simple Hamiltonian: $\hat{K}_0 = \sum_\alpha (\varepsilon_\alpha - \mu)\hat{a}_\alpha^\dagger \hat{a}_\alpha$

$$\left[\hat{a}_\alpha, \hat{a}_\beta^\dagger \hat{a}_\beta\right] = \delta_{\alpha\beta}\hat{a}_\beta \Rightarrow \left[\hat{a}_\alpha, \hat{K}_0\right] = (\varepsilon_\alpha - \mu_0)\hat{a}_\alpha \tag{3.38}$$

For Heisenberg picture:

$$i\hbar\frac{\partial}{\partial t}\hat{a}_\alpha(t) = \left[\hat{a}_\alpha(t), \hat{K}_0\right] = (\varepsilon_\alpha - \mu)\hat{a}_\alpha(t) \Rightarrow \hat{a}_\alpha(t) = e^{-i(\varepsilon_\alpha-\mu)(t-t_0)/\hbar}\hat{a}_\alpha(t_0) \tag{3.39}$$

$$i\hbar\frac{\partial}{\partial t}G_0(\alpha t;\alpha' t') = -ii\hbar\frac{\partial}{\partial t}\left[\theta(t-t')\left\langle\hat{a}_\alpha(t)\hat{a}_\alpha^\dagger(t')\right\rangle + \eta\theta(t'-t)\left\langle\hat{a}_\alpha^\dagger(t)\hat{a}_\alpha(t)\right\rangle\right] \tag{3.40}$$

For $t = t'$: $\frac{\partial}{\partial t}\theta(t-t') = \delta(t-t')$, for $t \neq t'$, focus on operator $i\hbar\frac{\partial}{\partial t}\hat{a}_\alpha(t) = (\varepsilon_\alpha - \mu)\hat{a}_\alpha(t)$:

$$\begin{aligned}\text{LHS} &= -i\delta(t-t')\left\langle\left[\hat{a}_\alpha(t)\hat{a}_\alpha^\dagger(t')\right]\right\rangle + (\varepsilon_\alpha - \mu)G_0(\alpha t;\alpha' t') \\ &= \hbar\delta_{\alpha\alpha'}\delta(t-t') + (\varepsilon_\alpha - \mu)G_0(\alpha t;\alpha' t')\end{aligned} \tag{3.41}$$

For fixed $\alpha', t'$, we treat it as a first-order linear differential equation on $t$. If we fix the diagonal property in $\alpha, \alpha'$ due to $\delta_{\alpha,\alpha'}$:

$$\begin{aligned}i\hbar\frac{\partial}{\partial t}G_\alpha(t,t') &= \hbar\delta(t-t') + (\varepsilon_\alpha - \mu)G_\alpha(t,t') \\ G_\alpha(t,t') &= C\exp(-i(\varepsilon_\alpha - \mu)(t-t')/\hbar)\end{aligned} \tag{3.42}$$

Where discontinuity condition:

$$i\hbar(G_\alpha(t-t'=0^+) - G_\alpha(t-t'=0^-)) = \hbar \Rightarrow G_\alpha(0^+) - G_\alpha(0^-) = -i \tag{3.43}$$

$$G_\alpha(\tau = t-t') = A_\pm \exp(-i(\varepsilon_\alpha - \mu)(t-t')/\hbar) \tag{3.44}$$

For $\tau > 0$ or $\tau < 0$.

$$\lim_{t\to t^+} G_0(\alpha t;\alpha' t') = -i\left\langle \hat{a}_\alpha(t')\hat{a}^\dagger_{\alpha'}(t')\right\rangle \tag{3.45}$$

Due to $\hat{K}_0 \propto \hat{a}^\dagger_\alpha \hat{a}_\alpha$ is diagonal, therefore each $\alpha, \alpha'$ belongs to different modes that:

$$\left\langle \hat{a}_\alpha \hat{a}^\dagger_{\alpha'}\right\rangle = \langle \hat{a}_\alpha\rangle \left\langle \hat{a}^\dagger_{\alpha'}\right\rangle = 0 \quad \alpha \neq \alpha' \tag{3.46}$$

$$\text{LHS} = -i\langle \hat{a}_\alpha(t')\hat{a}^\dagger_\alpha(t')\rangle = 1 + \eta n_\alpha, \quad n_\alpha = \langle \hat{a}^\dagger_\alpha \hat{a}_\alpha\rangle \tag{3.47}$$

Generally:

$$\lim_{t\to t^+} G_0(\alpha t;\alpha' t') = -i\delta_{\alpha\alpha'}(1+\eta n_\alpha), \quad \lim_{t\to t^-} G_0(\alpha t;\alpha' t') = -i\delta_{\alpha\alpha'}\eta n_\alpha \tag{3.48}$$

$$G_\alpha(0^+) = A_+ = -i(1+\eta n_\alpha), \quad G_\alpha(0^-) = A_- = -i\eta n_\alpha \tag{3.49}$$

$$G_0(\alpha t;\alpha' t') = -i\delta_{\alpha\alpha'}\exp(-i(\varepsilon_\alpha - \mu)(t-t'))[\theta(t-t')(1+\eta n_\alpha) + \theta(t'-t)\eta n_\alpha] \tag{3.50}$$

$$\tilde{G}_0(\alpha;\alpha')(\omega) = \int_{-\infty}^{\infty} dt\, G_0(\alpha t;\alpha' t') e^{-i\omega(t-t')} \tag{3.51}$$

Evaluate in separation:

$$\tilde{G}_0(\omega) = -i\left[(1+\eta n_\alpha)\int_0^\infty d\tau e^{i\omega\tau} e^{-i(\varepsilon_\alpha-\mu)\tau} + \eta n_\alpha \int_{-\infty}^0 d\tau e^{i\omega\tau} e^{-i(\varepsilon_\alpha-\mu)\tau}\right] \tag{3.52}$$

Here we translate $dt \to d(t-t') = d\tau$ where $t - t' > 0$ or $t - t' < 0$ for each part.

$$\begin{aligned}\tilde{G}_0(\omega) &= -i\left[(1+\eta n_\alpha)\frac{i}{\omega - \omega_\alpha + i0^+} + \eta n_\alpha \frac{-i}{\omega - \omega_\alpha - i0^+}\right] \\ &= \frac{1+\eta n_\alpha}{\omega - \omega_\alpha + i0^+} - \frac{\eta n_\alpha}{\omega - \omega_\alpha - i0^-}\end{aligned} \tag{3.53}$$

$$\tilde{G}_0(\alpha;\alpha')(\omega) = \delta_{\alpha\alpha'} e^{i\omega 0^+}\left[\frac{1+\eta n_\alpha}{\omega - (\varepsilon_\alpha - \mu) + i0^+} - \frac{\eta n_\alpha}{\omega - (\varepsilon_\alpha - \mu) - i0^-}\right] \tag{3.54}$$

We can define the dual of time and temperature: $t \leftrightarrow -i\tau$, $\tau \in [0, \hbar\beta) = [0, \hbar/kT)$

$$G^{(n)}(\alpha_i\tau_i, \alpha'_i\tau'_i) = -\left\langle \hat{T}\left[\prod_{i=1}^n \hat{a}^{(H)}_{\alpha_i}(\tau_i)\prod_{i=n}^1 \hat{a}^{\dagger(H)}_{\alpha'_i}(\tau'_i)\right]\right\rangle \tag{3.55}$$

$$\hat{a}^{(H)}_\alpha(\tau) = e^{\hat{H}\tau}\hat{a}_\alpha e^{-\hat{H}\tau} \tag{3.56}$$

To elucidate the periodicity, we omit any unnecessary symbol:

$$G(\tau) = -\left\langle \hat{T}\left[\hat{a}(\tau)\hat{a}^\dagger(0)\right]\right\rangle \tag{3.57}$$

$$G(\tau) = \begin{cases} -\langle \hat{a}(\tau)\hat{a}^\dagger(0)\rangle & \tau > 0 \\ -\eta\langle \hat{a}^\dagger(0)\hat{a}(\tau)\rangle & \tau < 0 \end{cases} \tag{3.58}$$

$$\begin{aligned} G(\beta^-) &= -\left\langle \hat{a}(\beta)\hat{a}^\dagger(0)\right\rangle = -\operatorname{tr}\left[e^{-\beta\hat{H}}e^{\beta H}\hat{a}e^{-\beta H}\hat{a}^\dagger\right] \\ &= -\operatorname{tr}\left[\hat{a}e^{-\beta\hat{H}}\hat{a}^\dagger\right] = -\operatorname{tr}\left[e^{-\beta\hat{H}}\hat{a}^\dagger\hat{a}\right] \\ &= \eta\, G(0^-) \end{aligned} \tag{3.59}$$

The $n$-point Green' s functions is same as below to take $\left[\prod_i A_i\right]$ as $\left[A_j X\right]$ by cyclic property of trace.

Assume the system is time-translation invariant, so the Green' s function depends only on the difference $\tau - \tau'$.

$$G(\tau) \coloneqq G(\alpha, \alpha'; \tau) \coloneqq G(\alpha\tau; \alpha' 0) \tag{3.60}$$

Where $\alpha, \alpha'$ could be omitted in cases we don' t pay attention.

$$G(\tau + \hbar\beta) = \eta\, G(\tau) \tag{3.61}$$

Suggests the discrete Fourier transformation, or Fourier series.

$$G(\tau) = \frac{1}{\hbar\beta}\sum_{-\infty}^{\infty} e^{-i\omega_n\tau}G(\omega_n) \tag{3.62}$$

$$\begin{aligned} e^{-i\omega_n(\tau+\hbar\beta)} &\equiv \eta\, e^{-i\omega_n\tau} \\ e^{-i\omega_n\hbar\beta} &= \eta 1 \\ \omega_n &= \begin{cases} \frac{2n\pi}{\hbar\beta} & \eta = 1 \\ \frac{(2n+1)\pi}{\hbar\beta} & \eta = -1 \end{cases} \end{aligned} \tag{3.63}$$

$$\int_0^{\hbar\beta} d\tau\, e^{i(\omega_n - \omega_m)\tau} \coloneqq \hbar\beta\delta_{nm} \tag{3.64}$$

$$G(\omega_n) = \int_0^{\hbar\beta} d\tau\, e^{i\omega_n\tau}G(\tau) \tag{3.65}$$

Restoring original notation:

$$G(\alpha\tau;\alpha'\tau') = \frac{1}{\hbar\beta}\sum_n e^{-i\omega_n(\tau-\tau')} G(\alpha;\alpha')(\omega_n) \tag{3.66}$$

$$G(\alpha;\alpha')(\omega_n) = \int_0^{\hbar\beta} d\tau\, e^{i\omega_n(\tau-\tau')} G(\alpha\tau;\alpha\tau') \tag{3.67}$$

For non-interacting system, the answer is simple that change all $(t-t') \leftrightarrow -i(\tau-\tau')$:

$$G_0(\alpha\tau,\alpha'\tau') = -\delta_{\alpha\alpha'}\exp(-(\varepsilon_\alpha-\mu)(\tau-\tau')/\hbar)[\theta(\tau-\tau')(1+\eta n_\alpha)+\eta\theta(\tau'-\tau)n_\alpha] \tag{3.68}$$

Suppose $\tau-\tau' \leftrightarrow \tau$:

$$\begin{aligned} G_0(0^-) &= G_0(\beta^-) \\ \eta n_\alpha &= \eta e^{-\beta(\varepsilon_\alpha-\mu)}(1+\eta n_\alpha) \\ n_\alpha\left(\eta - e^{-\beta(\varepsilon_\alpha-\mu)}\right) &= \eta e^{-\beta(\varepsilon_\alpha-\mu)} \\ n_\alpha &= \frac{1}{e^{\beta(\varepsilon_\alpha-\mu)}-\eta} \end{aligned} \tag{3.69}$$

$$G_0(\alpha,\alpha')(\omega_n) = \delta_{\alpha\alpha'}e^{i\omega 0^+}\left(\frac{1}{i\omega_n-(\varepsilon_\alpha-\mu)/\hbar}\right) \tag{3.70}$$

## 3.2 Continuum field operators

Kinetic energy:

$$\hat{T} = \int \mathrm{d}^3r\, \hat{\psi}^\dagger(\boldsymbol{r})\left(-\frac{\hbar^2}{2m}\nabla^2_{\boldsymbol{r}}\right)\hat{\psi}(\boldsymbol{r}) \tag{3.71}$$

Where $\hat{\psi}(\boldsymbol{r})$ is the field operator satisfying the equal time relation:

$$\left[\hat{\psi}(\boldsymbol{r}),\hat{\psi}^\dagger(\boldsymbol{r}')\right]_\mp = \delta(\boldsymbol{r}-\boldsymbol{r}') \tag{3.72}$$

The thermal expectation value is:

$$\begin{aligned} \left\langle\hat{T}\right\rangle &= \int \mathrm{d}^3r \left\langle \hat{\psi}^\dagger(\boldsymbol{r})\left(-\frac{\hbar^2}{2m}\right)\nabla^2_{\boldsymbol{r}}\hat{\psi}(\boldsymbol{r})\right\rangle \\ &= \int \mathrm{d}^3r \left\langle \hat{\psi}^\dagger(\boldsymbol{r}')\left(-\frac{\hbar^2}{2m}\right)\nabla^2_{\boldsymbol{r}}\hat{\psi}(\boldsymbol{r})\right\rangle\Bigg|_{\boldsymbol{r}'=\boldsymbol{r}} \\ &= \int \mathrm{d}^3r \left(-\frac{\hbar^2}{2m}\right)\nabla^2_{\boldsymbol{r}}\left\langle\hat{\psi}^\dagger(\boldsymbol{r}')\hat{\psi}(\boldsymbol{r})\right\rangle\Big|_{\boldsymbol{r}'=\boldsymbol{r}} \end{aligned} \tag{3.73}$$

Immediately, we inspect:[1]

$$
\begin{aligned}
G(\boldsymbol{r}t;\boldsymbol{r}'t') &= -i\left\langle \hat{T}\left[\hat{\psi}^{(H)}(\boldsymbol{r},t)\hat{\psi}^{\dagger(H)}(\boldsymbol{r}',t')\right]\right\rangle \\
i\eta\, G(\boldsymbol{r}t;\boldsymbol{r}'t^{+}) &= \left\langle \hat{\psi}^{(H)}(\boldsymbol{r},t)\hat{\psi}^{\dagger(H)}(\boldsymbol{r}',t^{+})\right\rangle
\end{aligned}
\tag{3.74}
$$

So:

$$
\left\langle \hat{T}\right\rangle = \int \mathrm{d}^3 r \left(-\frac{\hbar^2}{2m}\right)\nabla^2_{\boldsymbol{r}}[i\eta\, G(\boldsymbol{r}t;\boldsymbol{r}'t^{+})]\big|_{\boldsymbol{r}'=\boldsymbol{r}} \tag{3.75}
$$

Assume the system is translation invariant in space and time, so one can take: $\boldsymbol{R} = \boldsymbol{r} - \boldsymbol{r}', \tau = t - t'$

$$
G(\boldsymbol{R},\tau) \coloneqq G(\boldsymbol{r}t;\boldsymbol{r}'t') \tag{3.76}
$$

$$
\tilde{G}(\boldsymbol{k},\omega) = \int \mathrm{d}^3 R\, \mathrm{d}\tau e^{-i\boldsymbol{k}\cdot\boldsymbol{R}+i\omega\tau} G(\boldsymbol{R},\tau) \tag{3.77}
$$

$$
\nabla^2_{\boldsymbol{R}} G(0,0^-) = \int \frac{\mathrm{d}^3 k\, \mathrm{d}\omega}{(2\pi)^4}(i\boldsymbol{k})^2 e^{-i\omega 0^-}\tilde{G}(\boldsymbol{k},\omega)\Big|_{\boldsymbol{R}=0} = -\int \frac{\mathrm{d}^3 k\, \mathrm{d}\omega}{(2\pi)^4} k^2 e^{i\omega 0^+}\tilde{G}(\boldsymbol{k},\omega) \tag{3.78}
$$

$$
\begin{aligned}
\left\langle \hat{T}\right\rangle &= i\eta\mathcal{V}\left(-\frac{\hbar^2}{2m}\right)\left(-\int \frac{\mathrm{d}^3 k\, \mathrm{d}\omega}{(2\pi)^4} k^2 e^{i\omega 0^+}\tilde{G}(\boldsymbol{k},\omega)\right) \\
&= i\eta\mathcal{V}\int \frac{\mathrm{d}^3 k\, \mathrm{d}\omega}{(2\pi)^4} e^{i\omega 0^+}\frac{(\hbar k)^2}{2m}\tilde{G}(\boldsymbol{k},\omega)
\end{aligned}
\tag{3.79}
$$

Interaction energy:

It′ s simpler to deduce from:

$$
\begin{aligned}
\hat{\psi}^\dagger(\boldsymbol{r},t) i\hbar \frac{\partial}{\partial t}\hat{\psi}(\boldsymbol{r},t) &= \hat{\psi}^\dagger(\boldsymbol{r},t)\left[\hat{\psi}(\boldsymbol{r},t), \hat{T}+\hat{V}\right] \\
\hat{\psi}^\dagger(\boldsymbol{r},t) i\hbar \frac{\partial\hat{\psi}(\boldsymbol{r},t)}{\partial t} &= \hat{\psi}^\dagger(\boldsymbol{r},t)\left(-\frac{\hbar^2}{2m}\nabla^2 - \mu\right)\hat{\psi}(\boldsymbol{r},t) \\
&\quad + \int \mathrm{d}^3 r \hat{\psi}^\dagger(\boldsymbol{r},t)\hat{\psi}^\dagger(\boldsymbol{r}',t) v(\boldsymbol{r}-\boldsymbol{r}')\hat{\psi}(\boldsymbol{r}',t)\hat{\psi}(\boldsymbol{r},t)
\end{aligned}
\tag{3.80}
$$

[1]The $T$ is time operator now:

$$
\begin{aligned}
\left\langle \hat{V} \right\rangle &= \frac{i\eta}{2}\int \mathrm{d}^3 r \left[ \left( i\hbar\frac{\partial}{\partial t} + \frac{\hbar^2}{2m}\nabla_{\boldsymbol{r}}^2 + \mu \right) G(\boldsymbol{r}t;\boldsymbol{r}'t') \right] \Bigg|_{\boldsymbol{r}=\boldsymbol{r}',t'=t^+} \\
&= \frac{i\eta}{2}\mathcal{V}\int \frac{\mathrm{d}^3 k\,\mathrm{d}\omega}{(2\pi)^4} e^{i\omega 0^+} \left( \hbar\omega - \frac{(\hbar k)^2}{2m} + \mu \right) \tilde{G}(\boldsymbol{k},\omega)
\end{aligned}
\tag{3.81}
$$

## 3.3 Linear response

Linear Responses:

$$
\delta\hat{H} = -e\int \mathrm{d}\boldsymbol{r}\hat{\rho}(\boldsymbol{r})\phi(\boldsymbol{r}) \tag{3.82}
$$

Where $\phi(\boldsymbol{r})$ is external electric potential and $\hat{\rho}(\boldsymbol{r}) = \hat{\psi}^\dagger(\boldsymbol{r})\hat{\psi}(\boldsymbol{r})$ is the density operator.

$$
\hat{\jmath}(\boldsymbol{r}) = \frac{ie\hbar}{2m}\left[\hat{\psi}^\dagger(\boldsymbol{r})\left(\nabla\hat{\psi}(\boldsymbol{r})\right) - \left(\nabla\hat{\psi}^\dagger(\boldsymbol{r})\right)\hat{\psi}(\boldsymbol{r})\right] \tag{3.83}
$$

We only approximate in linear order, a merit is the total response is the summation of all field responses.

Consider a time-dependent infinitesimally small external field:

$$
\hat{H}_\phi(t) = \hat{H} + \hat{O}\phi(t) \tag{3.84}
$$

Where $\phi$ is the external field and $\hat{O}$ is the observable.

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}\hat{U}(t,t_i) &= \hat{H}_\phi(t)\hat{U}(t,t_i) \\
\hat{U}(t,t_i) &= \hat{T}\left[\exp\left(-\frac{i}{\hbar}\int_{t_i}^{t}\hat{H}_\phi(t)\,\mathrm{d}t\right)\right]
\end{aligned}
\tag{3.85}
$$

$$
\begin{aligned}
i\hbar\frac{\partial}{\partial t}\hat{U}(t,t_i) &= \left(\hat{H}+\hat{O}\phi(t)\right)\hat{U}(t,t_i), \quad \hat{U}(t_i,t_i)=\mathbf{1} \\
i\hbar\frac{\partial}{\partial t}\delta\hat{U}(t,t_i) &= \hat{H}\delta\hat{U}(t,t_i)+\hat{O}\phi(t)U_0(t,t_i)+O(\delta^2 U) \\
\hat{U}_0(t_i,t)i\hbar\frac{\partial}{\partial t}\delta\hat{U}(t,t_i)+i\hbar\frac{\partial}{\partial t}(U_0(t_i,t))\delta\hat{U}(t,t_i) &= U_0(t_i,t)\hat{O}\phi(t)U_0(t,t_i)+O(\delta^2 U) \\
i\hbar\frac{\partial}{\partial t}\left(U_0(t_i,t)\delta\hat{U}(t,t_i)\right) &= U_0(t_i,t)\hat{O}\phi(t)U_0(t,t_i)+O(\delta^2 U) \\
\delta\hat{U}(t,t_i) &= -\frac{i}{\hbar}\hat{U}_0(t,t_i)\int_{t_i}^{t}\mathrm{d}t_1\hat{U}_0(t_i,t_1)\hat{O}\phi(t_1)\hat{U}_0(t_1,t_i) \\
\delta\hat{U}(t,t_i) &= -\frac{i}{\hbar}\int_{t_i}^{t}\mathrm{d}t_1\hat{U}_0(t,t_1)\hat{O}\phi(t_1)\hat{U}_0(t_1,t_i)
\end{aligned}
\tag{3.86}
$$

Suppose $\hat{U}\approx\hat{U}_0+\delta\hat{U}$ where $\hat{U}_0$ is unperturbed evolution operator without $\hat{O}\phi(t)$, and **ignores** the second order.

$$
\delta|\psi(t)\rangle=\delta\hat{U}(t,t_i)|\psi(t_i)\rangle=\int_{t_i}^{t}\mathrm{d}t_1\frac{\delta\hat{U}(t,t_i)}{\delta\phi(t_1)}\phi(t_1)|\psi(t_i)\rangle \tag{3.87}
$$

$$
\begin{aligned}
\left.\frac{\delta\hat{U}(t,t_i)}{\delta\phi(t_1)}\right|_{\phi\to 0} &= -\frac{i}{\hbar}\hat{U}_0(t,t_1)\hat{O}\hat{U}_0(t_1,t_i), \quad t_i<t_1<t \\
&= -\frac{i}{\hbar}e^{-i\hat{H}(t-t_i)/\hbar}\left(e^{i\hat{H}(t_1-t_i)/\hbar}\hat{O}e^{-i\hat{H}(t_1-t_i)/\hbar}\right) \\
&= -\frac{i}{\hbar}e^{-i\hat{H}(t-t_i)}\hat{O}^{(H)}(t_1) \\
&= -\frac{i}{\hbar}U_0(t,t_i)\hat{O}^{(H)}(t_1)
\end{aligned}
\tag{3.88}
$$

$$
\mathrm{LHS}=-\frac{i}{\hbar}\int_{t_i}^{t}\mathrm{d}t_1 U_0(t,t_i)\hat{O}^{(H)}(t_1)\phi(t_1)|\psi(t_i)\rangle \tag{3.89}
$$

The expectation value of an observable $\hat{R}$ other than $\hat{O}$:

$$
\begin{aligned}
\delta\hat{R}(t) &= \delta\sum_n\rho_n\left\langle\psi_n(t)\middle|\hat{R}\middle|\psi_n(t)\right\rangle \\
&= \sum_n\rho_n\left(\left\langle\delta\psi_n(t)\middle|\hat{R}\middle|\psi_n(t)\right\rangle+\left\langle\psi_n(t)\middle|\hat{R}\middle|\delta\psi_n(t)\right\rangle\right)
\end{aligned}
\tag{3.90}
$$

Evaluate the first term:

$$
\begin{aligned}
\left\langle \delta\psi_n(t) \middle| \hat{R} \middle| \psi_n(t) \right\rangle &= \frac{i}{\hbar} \int_{t_i}^{t} \mathrm{d}t_1 \phi(t_1) \left\langle \psi_n(t_i) \middle| \hat{O}^{(H)}(t_1) U_0(t_i, t) \hat{R} \middle| \psi_n(t) \right\rangle \\
&= \frac{i}{\hbar} \int_{t_i}^{t} \mathrm{d}t_1 \phi(t_1) \left\langle \psi_n(t_i) \middle| \hat{O}^{(H)}(t_1) U_0(t_i, t) \hat{R} U_0(t, t_i) \middle| \psi_n(t_i) \right\rangle \\
&= \frac{i}{\hbar} \int_{t_i}^{t} \mathrm{d}t_1 \phi(t_1) \left\langle \psi_n(t_i) \middle| \hat{O}^{(H)}(t_1) \hat{R}^{(H)}(t) \middle| \psi_n(t_i) \right\rangle
\end{aligned} \tag{3.91}
$$

Therefore, suppose $t_i \to -\infty$ where we can compute the evolution by initial distribution:

$$
\mathrm{LHS} = -\frac{i}{\hbar} \int_{-\infty}^{\infty} \mathrm{d}t_1 \theta(t - t_1) \left\langle \left[ \hat{R}^{(H)}(t), \hat{O}^{(H)}(t_1) \right] \right\rangle \phi(t_1) \tag{3.92}
$$

$$
D^r\left(\hat{R}; \hat{O}\right)(t; t') = \frac{\mathrm{d}\hat{R}(t)}{\mathrm{d}\phi(t')} = -\frac{i}{\hbar} \theta(t - t') \left\langle \left[ \hat{R}^{(H)}(t), \hat{O}^{(H)}(t') \right] \right\rangle \tag{3.93}
$$

Scattering experiments:

$$
i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \left( \hat{H}_0 + \hat{V} \right) |\psi(t)\rangle \tag{3.94}
$$

We define interaction picture state as:

$$
|\psi_I(t)\rangle := e^{i\hat{H}_0 t/\hbar} |\psi(t)\rangle \tag{3.95}
$$

$$
\begin{aligned}
i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle &= \left( -\hat{H}_0 e^{i\hat{H}_0 t/\hbar} + e^{i\hat{H}_0 t/\hbar} \left( \hat{H}_0 + \hat{V} \right) \right) |\psi(t)\rangle \\
i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle &= e^{i\hat{H}_0 t/\hbar} \hat{V} e^{-i\hat{H}_0 t/\hbar} |\psi_I(t)\rangle \\
i\hbar \frac{\partial}{\partial t} |\psi_I(t)\rangle &= \hat{V}_I(t) |\psi_I(t)\rangle
\end{aligned} \tag{3.96}
$$

Start at $t = 0$ with $|\psi(0)\rangle = |\psi_i\rangle$. For a weak perturbation, we iterate to first order (Born approximation):

$$
|\psi_I(t)\rangle \approx |\psi_i\rangle - \frac{i}{\hbar} \int_0^t \mathrm{d}t' \hat{V}_I(t') |\psi_i\rangle \tag{3.97}
$$

The amplitude of finding the system in a different state $\psi_f$ at time $t$ is:

$$
\begin{aligned}
c_{fi}^{(1)}(t) = \langle \psi_f | \psi_i(t) \rangle &= -\frac{i}{\hbar} \int_0^t \mathrm{d}t' \left\langle \psi_f \middle| \hat{V}_I(t') \middle| \psi_i \right\rangle \\
&= -\frac{i}{\hbar} \int_0^t \mathrm{d}t' \left\langle \psi_f \middle| e^{i\hat{H}_0 t/\hbar} \hat{V} e^{-i\hat{H}_0 t/\hbar} \middle| \psi_i \right\rangle \\
&= -\frac{i}{\hbar} \int_0^t \mathrm{d}t' e^{i(E_f - E_i)/\hbar} \left\langle \psi_f \middle| \hat{V} \middle| \psi_i \right\rangle
\end{aligned} \tag{3.98}
$$

We determine the transition matrix from probed system $|\alpha\rangle$ to $|\beta\rangle$ and wave vector from $\boldsymbol{k}$ to $\boldsymbol{k} + \boldsymbol{q}$. Consider an external particle apply such effect on a many body system by scattering.

We prepare the initial state where:

- The many body system with eigenstate $|\alpha\rangle$ of $\hat{H}_{\text{sys}}$.
- The external particle is in a plane wave state $|\boldsymbol{k}\rangle$ with wave vector $\boldsymbol{k}$.

The initial state is:

$$
|\psi_i\rangle = |\boldsymbol{k}\rangle \otimes |\alpha\rangle \equiv |\boldsymbol{k}, \alpha\rangle \tag{3.99}
$$

The final state is:

$$
|\psi_f\rangle = |\boldsymbol{k} + \boldsymbol{q}\rangle \otimes |\beta\rangle \equiv |\boldsymbol{k} + \boldsymbol{q}, \beta\rangle \tag{3.100}
$$

$$
\hat{V} = \int \mathrm{d}^3 r \, \hat{\psi}^\dagger(\boldsymbol{r})_{\text{ext}} \hat{V}_{\text{ext}} \hat{\psi}_{\text{ext}}(\boldsymbol{r}) = \int \mathrm{d}^3 r \, \hat{\psi}^\dagger(\boldsymbol{r})_{\text{ext}} v(r - r_i) \hat{\psi}_{\text{ext}}(\boldsymbol{r}) \tag{3.101}
$$

$$
\left\langle \psi_f \middle| \hat{V} \middle| \psi_i \right\rangle = \langle \boldsymbol{k} + \boldsymbol{q}, \beta | \int \mathrm{d}^3 r \, \hat{\psi}^\dagger(\boldsymbol{r})_{\text{ext}} v(r - r_i) \hat{\psi}_{\text{ext}}(\boldsymbol{r}) | \boldsymbol{k}, \alpha\rangle \tag{3.102}
$$

Because:

$$
\hat{\psi}_{\text{ext}}(\boldsymbol{r}) |\boldsymbol{k}\rangle = \hat{\psi}_{\text{ext}}(\boldsymbol{r}) |\boldsymbol{r}\rangle \langle \boldsymbol{r} | \boldsymbol{k}\rangle \frac{1}{V^{\frac{1}{2}}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} |0\rangle_{\text{ext}} \tag{3.103}
$$

$$
\left\langle \psi_f \middle| \hat{V} \middle| \psi_i \right\rangle = T_{\alpha\beta}(\boldsymbol{q}) = \int \mathrm{d}^3 r \langle \beta | \left( e^{-i(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{r}} \sum_i v(\boldsymbol{r} - \boldsymbol{r}_i) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \right) |\alpha\rangle \tag{3.104}
$$

$$
\int \mathrm{d}^3 r e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} v(\boldsymbol{r} - \boldsymbol{r}_i) = e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} \int \mathrm{d}^3 R \, v(\boldsymbol{R}) = e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} \tilde{v}(\boldsymbol{q}) \tag{3.105}
$$

$$
T_{\alpha\beta}(\boldsymbol{q}) = \tilde{v}(\boldsymbol{q}) \left\langle \beta \middle| \sum_i e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} \middle| \alpha \right\rangle = \tilde{v}(\boldsymbol{q}) \langle \beta | \hat{\rho}(\boldsymbol{q}) | \alpha\rangle \tag{3.106}
$$

Where density matrix is $\hat{\rho}(\boldsymbol{q}) = \sum_i e^{-i\boldsymbol{q}\cdot\hat{\boldsymbol{r}}_i}$.

$$\begin{aligned} c^{(1)}_{\alpha\beta}(t) &= -\frac{i}{\hbar} T_{\alpha\beta}(\boldsymbol{q}) \int_0^t \mathrm{d}t' e^{i\omega_{\alpha\beta}}, \quad \omega_{\alpha\beta} = -\frac{E_\alpha - E_\beta}{\hbar} \\ &= -\frac{1}{\hbar} T_{\alpha\beta}(\boldsymbol{q}) \frac{e^{i\omega_{\alpha\beta}t} - 1}{\omega_{\alpha\beta}} \end{aligned} \tag{3.107}$$

$$P_{\alpha\beta}(\boldsymbol{q}, t) = \left|c^{(1)}_{\alpha\beta}(t)\right|^2 = \frac{\left|T_{\alpha\beta}(\boldsymbol{q})\right|^2}{\hbar^2} \frac{4\sin^2\left(\omega_{\alpha\beta}t/2\right)}{\omega^2_{\alpha\beta}} \tag{3.108}$$

We are interested in transitions to any final state $\beta$ of the many-body system, with the external particle′ s momentum transfer fixed to $\boldsymbol{q}$. The total probability for scattering with momentum transfer $q$ and energy transfer $\hbar\omega = E_\beta - E_\alpha$ is obtained by summing over all $\beta$. In a macroscopic system, the final states form a continuum; we introduce the density of states $\rho\left(E_\beta\right)$ at energy $E_\beta$. Then the total probability (for a given initial state $|\alpha\rangle$) is:

$$P_\alpha(\boldsymbol{q}, t) = \int \mathrm{d}E_\beta \rho\left(E_\beta\right) P_{\alpha\beta}(\boldsymbol{q}, t) \tag{3.109}$$

For large $t$, the function becomes sharply peaked around $E_\alpha = E_\beta$:

$$\frac{4\sin^2\left(\omega_{\alpha\beta}t/2\right)}{\omega^2_{\alpha\beta}} \approx 2\pi\hbar t \delta\left(E_\beta - E_\alpha\right) \tag{3.110}$$

$$P_\alpha(\boldsymbol{q}, t) \approx \sum_\beta \frac{2\pi t}{\hbar} \rho(E_\alpha) \left|T_{\alpha\beta}(\boldsymbol{q})\right|^2 \Big|_{E_\beta \to E_\alpha} \tag{3.111}$$

$$\Gamma_\alpha(\boldsymbol{q}) = \frac{\mathrm{d}P_\alpha(\boldsymbol{q}, t)}{\mathrm{d}t} = \sum_\beta \frac{2\pi}{\hbar} \rho(E_\alpha) \left|T_{\alpha\beta}(\boldsymbol{q})\right|^2 \Big|_{E_\beta \to E_\alpha} \tag{3.112}$$

In practise, the $E_\beta$ may differ $E_\alpha$ by a energy transfer $\hbar\omega$ we are interested in.

$$\begin{aligned} f\left(E_\beta - E_\alpha\right) &= \hbar \int d\omega f(\hbar\omega) \delta\left(E_\beta - E_\alpha - \hbar\omega\right) \\ \frac{\mathrm{d}}{\mathrm{d}\omega} f\left(E_\beta - E_\alpha\right) &= \hbar f(\hbar\omega) \delta\left(E_\beta - E_\alpha - \hbar\omega\right) \end{aligned} \tag{3.113}$$

$$\sigma_{\alpha(\boldsymbol{q},\omega)} = \frac{\mathrm{d}}{\mathrm{d}\omega} \Gamma_\alpha(\boldsymbol{q})(\omega) = 2\pi \sum_\beta \rho(E_\alpha) \delta\left(E_\beta - E_\alpha - \hbar\omega\right) \left|T_{\alpha\beta}(\boldsymbol{q})\right|^2 \tag{3.114}$$

$$\delta\big(E_\beta - E_\alpha - \hbar\omega\big) = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} \mathrm{d}t e^{i(E_\beta - E_\alpha - \hbar\omega)t/\hbar} \tag{3.115}$$

$$\begin{aligned}\sum_\beta e^{i(E_\beta - E_\alpha)t/\hbar}\langle\alpha|\hat{\rho}^\dagger(\boldsymbol{q})|\beta\rangle\langle\beta|\hat{\rho}(\boldsymbol{q})|\alpha\rangle &= \left\langle\alpha\middle|e^{i\hat{H}t/\hbar}\hat{\rho}^\dagger(\boldsymbol{q})e^{-i\hat{H}t/\hbar}\hat{\rho}(\boldsymbol{q})\middle|\alpha\right\rangle \\ &= \left\langle\alpha\middle|\hat{\rho}^\dagger(\boldsymbol{q})(t)\hat{\rho}\middle|\alpha\right\rangle\end{aligned} \tag{3.116}$$

$$\sigma_\alpha(\boldsymbol{q},\omega) \propto |\tilde{v}(\boldsymbol{q})|^2\rho(E_\alpha)\int \mathrm{d}t\left\langle\alpha\middle|\hat{\rho}^\dagger(\boldsymbol{q})(t)\hat{\rho}\middle|\alpha\right\rangle \tag{3.117}$$

## 3.4 Real-time Green functions

$$G(rt;r't') = -i\left\langle\hat{T}\left[\hat{\psi}(r,t)\hat{\psi}^\dagger(r',t')\right]\right\rangle \tag{3.118}$$

We omit irrelevant indices, coordinates and focus on Green’ s function:

$$\begin{aligned}G &= -i\left\langle\hat{T}\left[\psi\psi^\dagger\right]\right\rangle = -i\left[\theta(t-t')\left\langle\psi\psi^\dagger\right\rangle + \eta\theta(t'-t)\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= \theta(t-t')\left(-i\left\langle\psi\psi^\dagger\right\rangle\right) + \theta(t'-t)\left(-i\eta\left\langle\psi^\dagger\psi\right\rangle\right) \\ &= \theta(t-t')G^> + \theta(t'-t)G^<\end{aligned} \tag{3.119}$$

Another way to decompose is by identity:

$$\theta(t-t') + \theta(t'-t) = \mathbf{1} \tag{3.120}$$

$$\begin{aligned}G &= -i\left[\theta(t-t')\left\langle\psi\psi^\dagger\right\rangle + \eta\theta(t'-t)\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= -i\left[\theta(t-t')\left\langle\psi\psi^\dagger\right\rangle + \eta(1-\theta(t-t'))\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= -i\left[\theta(t-t')\left\langle\left[\psi,\psi^\dagger\right]\right\rangle + \eta\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= G^r + G^<\end{aligned} \tag{3.121}$$

$$\begin{aligned}G &= -i\left[\theta(t-t')\left\langle\psi\psi^\dagger\right\rangle + \eta\theta(t'-t)\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= -i\left[(1-\theta(t'-t))\left\langle\psi\psi^\dagger\right\rangle + \eta\theta(t'-t)\left\langle\psi^\dagger\psi\right\rangle\right] \\ &= -i\left[\left\langle\psi\psi^\dagger\right\rangle - \theta(t'-t)\left\langle\left[\psi,\psi^\dagger\right]\right\rangle\right] \\ &= G^> + G^a\end{aligned} \tag{3.122}$$

$$G^r + G^< = G^> + G^a \Rightarrow G^r - G^a = G^> - G^< \tag{3.123}$$

We assume $\langle O\rangle^* = \left\langle O^\dagger\right\rangle$, which is true for thermal state or ground states with Hermitian Hamiltonian.

$$[A,B]^\dagger = \left[B^\dagger, A^\dagger\right] \Rightarrow \left[\psi(\boldsymbol{r},t), \psi^\dagger(\boldsymbol{r}',t')\right]^\dagger = \left[\psi(\boldsymbol{r}',t'), \psi^\dagger(\boldsymbol{r},t)\right] \tag{3.124}$$

$$\begin{aligned}
(G^r)^*(rt;r't') &= \left(-i\theta(t-t')\left\langle\left[\psi,\psi^\dagger\right]\right\rangle\right)^* = i\theta(t-t')\left\langle\left[\psi,\psi^\dagger\right]^\dagger\right\rangle \\
&= i\theta(t-t')\left\langle\left[\psi(\boldsymbol{r}',t'),\psi^\dagger(\boldsymbol{r},t)\right]\right\rangle \\
&\overset{t\leftrightarrow t'}{=} G^a(r't';rt)
\end{aligned} \tag{3.125}$$

Reversely:

$$(G^a)^*(rt;r't') = G^r(r't';rt) \tag{3.126}$$

Assume translation invariance in space and time that $t \to t - t'$, $r \to r - r'$:

$$G^r(\boldsymbol{k},\omega) = -i\int_{-\infty}^{\infty} \mathrm{d}^3r \int_{-\infty}^{\infty} \mathrm{d}t e^{-i\boldsymbol{k}\cdot\boldsymbol{r}+i\omega t}\, G^r(\boldsymbol{r}t;\boldsymbol{r}'t') \tag{3.127}$$

Then, indeed:

$$\begin{aligned}
A(\boldsymbol{k},\omega) &= i(G^r(\boldsymbol{k},\omega) - G^a(\boldsymbol{k},\omega)) = i(G^r(\boldsymbol{k},\omega) - (G^r)^*(\boldsymbol{k},\omega)) \\
&= i\cdot 2i\,\mathrm{Im}\,G^r(\boldsymbol{k},\omega) = -2\,\mathrm{Im}\,G^r(\boldsymbol{k},\omega)
\end{aligned} \tag{3.128}$$

It′ s same that: $A(\boldsymbol{k},\omega) = i(G^> - G^<)$ too.

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} A(\boldsymbol{k},\omega) &= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}t\, e^{i\omega t}(...) \\
&= \int_{-\infty}^{\infty} \delta(t)\, G(t,\boldsymbol{k}) = G(0,\boldsymbol{k})
\end{aligned} \tag{3.129}$$

Notice that when $t \leftrightarrow t - t' = 0$, where the system at initial state has the canonical commutation relation, therefore:

$$\left\langle\left[\psi,\psi^\dagger\right]\right\rangle = 1 \Rightarrow G(0,\boldsymbol{k}) = 1 \tag{3.130}$$

Then the spectrum identity is obvious:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} A(\boldsymbol{k},\omega) = 1 \tag{3.131}$$

The spectrum in momentum space is more clear:

$$A(\boldsymbol{k},\omega) = \int_{-\infty}^{\infty} \mathrm{d}t e^{i\omega t}\left\langle\left[\hat{a}_k(t), \hat{a}_k^\dagger\right]\right\rangle \tag{3.132}$$

We can check by non-interacting system:

$$\hat{a}_k(t) = e^{-i(\varepsilon-\mu)t}\hat{a}_k \Rightarrow \left[\hat{a}_k(t), \hat{a}_k^\dagger\right] = e^{-i(\varepsilon_k-\mu)t} \tag{3.133}$$

$$A(\boldsymbol{k},\omega) = \int_{-\infty}^{\infty} \mathrm{d}t e^{i\omega t} e^{i(\varepsilon-\mu)t} = 2\pi\delta(\omega - (\varepsilon_k - \mu)) \tag{3.134}$$

## 3.5 Spectral representation

$$G^<(\boldsymbol{k},t) = -i\eta \sum_{m,n} e^{-\beta K_n} |\langle m|\hat{a}_k|n\rangle|^2 e^{i(K_m-K_n)t/\hbar} \tag{3.135}$$

$$G^<(\boldsymbol{k},\omega) = \int \mathrm{d}t e^{i\omega t} G^<(\boldsymbol{k},t) = -i\eta \sum_{n,m} e^{-\beta K_n} 2\pi\delta(\omega - (K_n - K_m)/\hbar)|\langle m|\hat{a}_k|n\rangle|^2 \tag{3.136}$$

Similarly:

$$G^>(\boldsymbol{k},t) = -i \sum_{n,m} e^{-\beta K_n} |\langle m|\hat{a}_k|n\rangle|^2 e^{i(K_n-K_m)t/\hbar} \tag{3.137}$$

To build the connection between $G^>$ and $G^<$, we relabel $n \leftrightarrow m$ for consistency:

$$G^>(\boldsymbol{k},\omega) = -i \sum_{n,m} e^{-\beta K_m} 2\pi\delta(\omega - (K_n - K_m)/\hbar)|\langle m|\hat{a}_k|n\rangle|^2 \tag{3.138}$$

$$G^<(\boldsymbol{k},\omega) = \eta e^{-\beta(K_n-K_m)} G^>(\boldsymbol{k},\omega) = \eta e^{-\beta\hbar\omega} G^>(\boldsymbol{k},\omega) \tag{3.139}$$

$$G^>(\boldsymbol{k},\omega) = \eta e^{\beta\hbar\omega} G^<(\boldsymbol{k},\omega) \tag{3.140}$$

$$A(\boldsymbol{k},\omega) = i[G^>(\boldsymbol{k},\omega) - G^<(\boldsymbol{k},\omega)] = i(G^>(\boldsymbol{k},\omega)\left(\eta e^{\beta\hbar\omega} - 1\right) \tag{3.141}$$

$$G^<(\boldsymbol{k},\omega) = -i\eta \frac{A}{e^{\beta\hbar\omega} - \eta} \Rightarrow G^< = -i\eta n_\eta(\omega) A(\boldsymbol{k},\omega) \tag{3.142}$$

$$G^>(\boldsymbol{k},\omega) = -i\eta e^{\beta\hbar\omega} \cdot n_\eta(\omega) A(\boldsymbol{k},\omega) = -i\left(1 + \eta n_\eta\right) A(\boldsymbol{k},\omega) \tag{3.143}$$

$$\begin{aligned} G^>(\boldsymbol{k},t) &= -i \int \mathrm{d}\omega_1 e^{-i\omega_1 t} \left[1 + \eta n_\eta(\omega_1)\right] A(\boldsymbol{k},\omega) \\ G^<(\boldsymbol{k},t) &= -i\eta \int \mathrm{d}\omega_1 e^{-i\omega_1 t} n_\eta(\omega_1) A(\boldsymbol{k},\omega) \end{aligned} \tag{3.144}$$

The expression by $A(\boldsymbol{k},\omega)$ can also be applied to $G^r$ and $G^a$:

$$
\begin{aligned}
G^r(\boldsymbol{k},t) &= \theta(t)(G^>(\boldsymbol{k},t) - G^<(\boldsymbol{k},t)) \\
G^a(\boldsymbol{k},t) &= -\theta(-t)(G^>(\boldsymbol{k},t) - G^<(\boldsymbol{k},t))
\end{aligned}
\tag{3.145}
$$

Because:

$$
A(\boldsymbol{k},\omega) = i[G^>(\boldsymbol{k},\omega) - G^<(\boldsymbol{k},\omega)]
\tag{3.146}
$$

$$
\begin{aligned}
G^r(\boldsymbol{k},\omega) = \int \mathrm{d}t e^{i\omega t}\theta(t)(-i)\int \frac{\mathrm{d}\omega_1}{2\pi} e^{-i\omega_1 t} A = -i\int \frac{\mathrm{d}\omega_1}{2\pi} A(\boldsymbol{k},\omega_1)\int_0^\infty e^{i(\omega-\omega_1)t} \\
= -i\cdot i\int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega_1)}{\omega-\omega_1+i0^+} = \int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega_1)}{\omega-\omega_1+i0^+}
\end{aligned}
\tag{3.147}
$$

$$
G^a(\boldsymbol{k},\omega) = -(-i)\int \frac{\mathrm{d}\omega_1}{2\pi} A(\boldsymbol{k},\omega_1)\int_{-\infty}^0 \mathrm{d}t e^{i(\omega-\omega_1)t} = \int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega_1)}{\omega-\omega_1-i0^+}
\tag{3.148}
$$

$$
\begin{aligned}
G(\boldsymbol{k},t) = \theta(t)G^>(\boldsymbol{k},t) + \theta(-t)G^<(\boldsymbol{k},t) = \int \mathrm{d}t e^{i\omega t}[\theta(t)G^>(\boldsymbol{k},\omega) + \theta(-t)G^<(\boldsymbol{k},\omega)] \\
= \int \frac{\mathrm{d}\omega_1}{2\pi} A\left[\frac{1+\eta n_\eta(\omega)}{\omega-\omega_1+i0^+} - \frac{\eta n_\eta(\omega)}{\omega-\omega_1-i0^+}\right]
\end{aligned}
\tag{3.149}
$$

A useful identity can be applied to separate real and imaginary part:

$$
\frac{1}{x\pm i0^+} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x)
\tag{3.150}
$$

$$
G^r(\boldsymbol{k},\omega) = \int \frac{\mathrm{d}\omega_1}{2\pi} A\left[\mathcal{P}\left(\frac{1}{\omega-\omega_1} - i\pi\delta(\omega-\omega_1)\right)\right] = \mathcal{P}\left(\int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega)}{\omega-\omega_1}\right) - \frac{i}{2}A(\boldsymbol{k},\omega)
\tag{3.151}
$$

$$
\operatorname{Re} G^r = \mathcal{P}\left(\int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega)}{\omega-\omega_1}\right), \quad \operatorname{Im} G^r = -\frac{1}{2}A
\tag{3.152}
$$

$$
\operatorname{Re} G^a = \operatorname{Re} G^r = \mathcal{P}\left(\int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega)}{\omega-\omega_1}\right), \quad \operatorname{Im} G^a = \frac{1}{2}A
\tag{3.153}
$$

$$
\begin{aligned}
G &= \int \frac{\mathrm{d}\omega_1}{2\pi} A\left\{1 + \eta n_\eta\left[\mathcal{P}\left(\frac{1}{\omega-\omega_1}\right) - i\pi\delta(\omega-\omega_1)\right] + \dots\right\} \\
&= \mathcal{P}\left(\int \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega)}{\omega-\omega_1}\right) - i\pi\int \frac{\mathrm{d}\omega_1}{2\pi} A\,\delta(\omega-\omega_1)\big(1+\eta n_\eta + \eta n_\eta\big)
\end{aligned}
\tag{3.154}
$$

$$
\operatorname{Im} G = -\frac{1}{2}\big(1 + 2\eta n_\eta(\omega)\big)A(\boldsymbol{k},\omega)
\tag{3.155}
$$

Interestingly, the particle density distribution can be further reduced:

$$1 + 2\eta n_\eta(\omega) = 1 + \frac{2\eta}{e^{\beta\hbar\omega} - \eta} = \frac{e^{\beta\hbar\omega} + \eta}{e^{\beta\hbar\omega} - \eta} = \frac{e^{\beta\hbar\omega/2} + \eta e^{-\beta\hbar\omega/2}}{e^{\beta\hbar\omega/2} - \eta e^{-\beta\hbar\omega/2}} = \left(\tanh\left(\frac{\beta\hbar\omega}{2}\right)\right)^{-\eta} \tag{3.156}$$

$$\operatorname{Im} G = -\frac{1}{2}\tanh^{-\eta}\left(\frac{\beta\hbar\omega}{2}\right)A(\boldsymbol{k},\omega) \tag{3.157}$$

Generally, one can use analytic continuation to acquire all Green' s functions:

$$G(\boldsymbol{k},z) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}\frac{A(\boldsymbol{k},\omega)}{z-\omega} \tag{3.158}$$

$$G^r(\boldsymbol{k},\omega) = \lim_{\varepsilon\to 0^+} G(k,\omega - i\eta) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega)}{\omega-\omega_1 - i\varepsilon} \tag{3.159}$$

Indeed:

$$G^a(\boldsymbol{k},\omega) = \lim_{\varepsilon\to 0^+} G(k,\omega - i\varepsilon) \tag{3.160}$$

$$G(\boldsymbol{k},\tau = \omega_n) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega_1}{2\pi}\frac{A(\boldsymbol{k},\omega_1)}{i\omega_n - \omega_1} \tag{3.161}$$

## 3.6 Nonequilibrium evolution

$$\hat{U}(t,t') = \hat{T}\exp\left(-\frac{i}{\hbar}\int_{t'}^{t}\hat{H}(\tau)\,\mathrm{d}\tau\right) \tag{3.162}$$

With convention $\hat{U}(t,t') = \hat{U}^\dagger(t',t)$.

$$\begin{aligned} G^>(rt;r't') &= -i\left\langle \hat{\psi}^{(H)}(r,t)\hat{\psi}^{\dagger(H)}(r',t')\right\rangle \\ &= -i\operatorname{tr}\left[e^{-\beta\hat{H}(t_i)}\hat{U}(t_i,t)\hat{\psi}(r)\hat{U}(t,t')\hat{\psi}^\dagger(r')\hat{U}(t',t_i)\right] \end{aligned} \tag{3.163}$$

Where $t_i$ is the initial time. Now we rewrite the thermal factor as a evolution along the imaginary axis:

$$e^{-\beta\hat{H}(t_i)} = \hat{U}(t_i - i\hbar\beta, t_i) \tag{3.164}$$

$$G^>(rt;r't') = -i\operatorname{tr}\left[\hat{U}(t_i - i\hbar\beta,t_i)\hat{U}(t_i,t)\hat{\psi}(r)\hat{U}(t,t')\hat{\psi}^\dagger(r')\hat{U}(t',t_i)\right] \tag{3.165}$$

We can formalize the specific contour that from $t_i^+ \to t'^+ \to t^- \to t_i^- \to t_i^- - i\hbar\beta$ that from above real axis to below real axis and down to imaginary axis, we assign the above, below and down to imaginary one each as $C^+, C^-, C^\beta$.

# 4 Path integrals

## 4.1 Time slicing

$$\hat{U}(t_f, t_i) = e^{-\frac{i}{\hbar}\hat{H}(t_f - t_i)} \tag{4.166}$$

$$e^{\hat{A}+\hat{B}} = \lim_{M\to\infty} \left(e^{\hat{A}/M} e^{\hat{B}/M}\right)^M \tag{4.167}$$

$$e^{-\frac{i}{\hbar}\hat{H}(t_f - t_i)} = \lim_{M\to\infty} \left(e^{-\frac{i}{\hbar}\hat{T}\varepsilon} e^{-\frac{i}{\hbar}\hat{V}\varepsilon}\right)^M \tag{4.168}$$

Where $\varepsilon = (t_f - t_i)/M$.

$$U(x_f, t_f; x_i, t_i) = \lim_{M\to\infty} \left\langle x_f \middle| \left(e^{-\frac{i}{\hbar}\hat{T}\varepsilon} e^{-\frac{i}{\hbar}\hat{V}\varepsilon}\right)^M \middle| x_i \right\rangle \tag{4.169}$$

$$U = \lim_{M\to\infty} \int \prod_{k=1}^{M-1} \mathrm{d}x_k \left\langle x_f \middle| e^{-\frac{i}{\hbar}\hat{T}\varepsilon} e^{-\frac{i}{\hbar}\hat{V}\varepsilon} \middle| x_{M-1} \right\rangle \cdots \tag{4.170}$$

$$\left\langle x_n \middle| e^{-\frac{i}{\hbar}\hat{T}\varepsilon} e^{-\frac{i}{\hbar}\hat{V}\varepsilon} \middle| x_{n-1} \right\rangle = \int \frac{\mathrm{d}p_n}{2\pi\hbar} e^{\frac{i}{\hbar}p_n(x_n - x_{n-1})} e^{-\frac{i}{\hbar}\frac{p_n^2}{2m}\varepsilon} e^{-\frac{i}{\hbar}V(x_{n-1})\varepsilon} \tag{4.171}$$

$$\begin{aligned} U = \lim_{M\to\infty} \int \prod_{k=1}^{M-1} \mathrm{d}x_k \prod_{k=1}^{M} \frac{\mathrm{d}p_k}{2\pi\hbar} \\ \exp\left(\frac{i}{\hbar} \sum_{k=1}^{M} (p_k(x_k - x_{k-1})) - \varepsilon H(p_k, x_{k-1})\right) \end{aligned} \tag{4.172}$$

$$x_k, k \in \mathbb{N} \Rightarrow x(k), k \in \mathbb{N} \overset{k\to\infty}{\Rightarrow} x(t), t \in \mathbb{R} \tag{4.173}$$

$$d\mu(x) = \prod_{k=1}^{M-1} \mathrm{d}x_k \overset{M\to\infty}{\Rightarrow} \mathrm{D}x(t) \tag{4.174}$$

$$U(x_f, t_f; x_i, t_i) = \int_{x(t_i)=t_i}^{x(t_f)=x_f} \mathrm{D}x(t) \int_{-\infty}^{\infty} \mathrm{D}p(t) \exp\left(\frac{i}{\hbar}\, \mathrm{d}t L(p, x)(t)\right) \tag{4.175}$$

$$L(x,\dot{x})(t) = p(t)\dot{x}(t) - H(p(t), x(t)) \tag{4.176}$$

$$S[x(t)] = \int_{x_i}^{x_f} L(x,\dot{x})(t) \tag{4.177}$$

If the kinematic energy is the simple form $p^2/2m$, we can integrate it and therefore:

$$\mathrm{D}x(t) = \lim_{M\to\infty} \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{\frac{M}{2}} \prod_{k=1}^{M-1} \frac{\mathrm{d}x_k}{\sqrt{\text{scale}}} \tag{4.178}$$

For $t_1 > t_2$, we start with Heisenberg picture operators:

$$\hat{O}^{(H)}(t) = e^{\frac{i}{\hbar}\hat{H}(t-t_i)}\hat{O}e^{-\frac{i}{\hbar}\hat{H}(t-t_i)} \tag{4.179}$$

The time ordered product is:

$$\hat{T}\left(\hat{O}^{(H)}(t_1)\hat{O}^{(H)}(t_2)\right) = \hat{O}^{(H)}(t_1)\hat{O}^{(H)}(t_2) \quad t_1 > t_2 \tag{4.180}$$

$$\left\langle x_f \middle| T\left[\hat{O}^{(H)}(t_1), \hat{O}^{(H)}(t_2)\right] \middle| x_i \right\rangle = \int \mathrm{D}x(t) O(x(t_1)) O(x(t_2)) e^{\frac{i}{\hbar}S[x(t)]} \tag{4.181}$$

$$Z = \operatorname{tr} e^{-\beta\hat{H}} = \int_{x(\tau_i)=x_0}^{x(\tau_f)=x_0} \left\langle x \middle| e^{-\beta\hat{H}} \middle| x \right\rangle \tag{4.182}$$

Using the same time-slicing idea, with $\varepsilon = \hbar\beta/M$, $\tau_f - \tau_i = \hbar\beta$.

$$U(x_f, \tau_f; x_i, \tau_i) = \left\langle x_f \middle| e^{-\frac{1}{\hbar}(\tau_f - \tau_i)\hat{H}} \middle| x_i \right\rangle, \quad \tau = it \tag{4.183}$$

Replace with $t = -i\tau$:

$$Z = \int_{x(\hbar\beta)=x(0)} \mathrm{D}x(\tau) \exp\left(-\frac{1}{\hbar} S_E[x(\tau)]\right) \tag{4.184}$$

$$\begin{aligned} S_E[x(\tau)] &= iS(x(-i\tau))|_{x(\hbar\beta)=x(0)} \\ &= \frac{i}{\hbar} \int_{x(\hbar\beta)=x(0)} -i\,\mathrm{d}\tau \left(\frac{m}{2}\frac{\mathrm{d}^2 x}{\mathrm{d}(-i\tau)^2} + V(x)\right) \\ &= \frac{1}{\hbar} \int_{x(\hbar\beta)=x(0)} \mathrm{d}\tau \left(\frac{m}{2}\frac{\mathrm{d}^2 x}{\mathrm{d}\tau^2} + V(x)\right) \end{aligned} \tag{4.185}$$

For time-slicing $k$, we can build upon the classical interpretation:

$$H(\{x_k\}) = \frac{1}{\hbar\beta}\sum_{k=1}^{M}\left[\frac{m}{2}\frac{(x_k - x_{k-1})^2}{\varepsilon^2} + V(x_{k-1})\right]\varepsilon \tag{4.186}$$

Is generally a harmonic string where potential is split equally among these beads.

$$Z = \frac{1}{N!}\sum_{\sigma}\eta^{|\sigma|}\int\prod_{i=1}^{N}\mathrm{D}x_i(\tau)\delta\Big(x_i(\hbar\beta) - x_{\sigma(i)}(0)\Big)e^{-S_E/\hbar} \tag{4.187}$$

For all permutation $\sigma$, where $\eta = +1$ for bosons and $-1$ for fermions.

### 4.1.1 Free Particle

$$Z_{\text{free}} = \int_{x(\hbar\beta)=x(0)}\mathrm{d}x\,\exp\left(-\frac{m}{2\hbar}\int_0^{\hbar\beta}\mathrm{d}\tau\dot{x}^2\right) \tag{4.188}$$

Because $x(\tau)$ satisfy boundary condition as a fourier modes:

$$x(\tau) = \sum_{n=-\infty}^{\infty}x_n e^{i\omega_n\tau},\quad \omega_n = \frac{2\pi n}{\hbar\beta},\quad x_{-n} = x_n^* \tag{4.189}$$

$$\begin{aligned}\int_0^{\hbar\beta}\mathrm{d}\tau\dot{x}^2 &= \sum_{n=-\infty}^{\infty}\sum_{n'=-\infty}^{\infty}\int_0^{\hbar\beta}\mathrm{d}\tau\, i\omega_n x_n i\omega_{n'}x_{n'}e^{i(\omega_n-\omega_{n'})\tau} \\ &= \sum_{n=-\infty}^{\infty}\sum_{n'=-\infty}\hbar\beta\delta_{nn'}x_n x_{n'}\omega_n\omega_{n'} \\ &= \hbar\beta\sum_{n=-\infty}^{\infty}|x_n|^2\end{aligned} \tag{4.190}$$

The mesasure $\mathrm{D}x$ is proportional to $\prod_n \mathrm{d}x_n\,\mathrm{d}x_{-n}$ with appropriate normalization.

$$Z_{\text{free}} = \prod_{n\geq 0}\int\mathrm{d}x_n\,\mathrm{d}x_{-n}\exp\Big(-\frac{m}{2\hbar}\hbar\beta\omega_n^2\Big(|x_n|^2 + |x_{-n}|\Big)\Big) \tag{4.191}$$

$$\int\mathrm{d}x_n\exp\Big(-\frac{m}{2\hbar}\hbar\beta\omega_n^2|x_n|^2\Big) = \left(2\frac{\pi}{m\beta\omega_n^2}\right)^{\frac{1}{2}} \tag{4.192}$$

For zero eigenvlue $x(\tau) = \text{const}$:

$$x(\tau) = x_0 + x'(\tau) \Rightarrow Z[x_0] = \int_{-\infty}^{\infty}\mathrm{d}x_0 = V \tag{4.193}$$

Then

$$\frac{1}{\hbar}\hbar\beta\omega_n^2 = \frac{(2\pi n)^2}{\hbar^2\beta} \tag{4.194}$$

$$Z_{\text{free}} = V\prod_{n=1}^{\infty}\left(\frac{\hbar^2\beta}{2\pi m n^2}\right) \tag{4.195}$$

Define the spectral zeta function $\zeta_\Pi(s)$ associated with the iegenvalues $\lambda_n$:

$$\prod_{n=1}^{\infty}\lambda_n = \exp\left(\sum_{n=1}^{\infty}\ln\lambda_n\right) \tag{4.196}$$

$$\sum_{n=1}^{\infty}\lambda_n^{-s}\ln\lambda_n = \left(-\sum_{n=1}^{\infty}\lambda_n^{-s}\right)\Bigg|_{s=0} = -\zeta_\Pi'(s=0) \tag{4.197}$$

$$\zeta_\Pi(s) = \sum_{n=1}^{\infty}\lambda_n^{-s} \tag{4.198}$$

So it equal to:

$$Z_{\text{free}} = V\exp\left(-\zeta\left(\frac{\hbar^2\beta}{(2\pi m n^2)'}\right)(s=0)\right) \tag{4.199}$$

Generally, for $\zeta(s) = \sum_{n\geq 1}\frac{1}{n^s}$:

$$\zeta(0) = -\frac{1}{2} \tag{4.200}$$

$$\zeta'(0) = \frac{1}{2}\ln(2\pi) \tag{4.201}$$

Which we will not prove here.

Suppose $\lambda_n = cn^{-s}$:

$$\zeta_\Pi'(0) = -\frac{1}{2}\ln(c) + \ln(2\pi) \tag{4.202}$$

$$-\zeta_\Pi'(0) = \frac{1}{2}\ln(c) - \ln(2\pi) = \ln(\left(\frac{m}{(2\pi\hbar^2\beta)^{\frac{1}{2}}}\right), \quad c = \frac{2\pi m}{\hbar^2\beta} \tag{4.203}$$

$$Z_{\text{free}} = V\left(\frac{m}{2\pi\hbar^2\beta}\right)^{\frac{1}{2}} \tag{4.204}$$

This is exactly the well-known result for a free particle in 1-dimension for a volume $V$

### 4.1.2 Vector Potential

For particle in vector potential, the action, which is Lagrange of particle in vector potential is:

$$S[\boldsymbol{x}] = \int_{x_i}^{x_f} \mathrm{d}t[L(x, \dot{x}) - e\boldsymbol{A}(\boldsymbol{x}) \cdot \dot{\boldsymbol{x}}] \tag{4.205}$$

For certain Lagrange of the particle without vector potential.

The Euclidean action is the variable transformation:

$$S_E[\boldsymbol{x}] = \int_0^{\hbar\beta} \mathrm{d}\tau[L(x, \dot{x}) + ie\boldsymbol{A}(\boldsymbol{x}) \cdot \dot{\boldsymbol{x}}] \tag{4.206}$$

We focus on the latter term:

$$\int_0^{\hbar\beta} \mathrm{d}\tau ie\boldsymbol{A}(\boldsymbol{x}) \cdot \dot{\boldsymbol{x}} = ie \int_{x_i}^{x_f} \mathrm{d}\boldsymbol{x} \cdot \boldsymbol{A} \tag{4.207}$$

Which is the line integral of vector potential along the path. If path closed:

$$\mathrm{LHS} = ie \int_S \mathrm{d}\boldsymbol{S} \cdot (\nabla \times A) = ie \int_S \mathrm{d}\boldsymbol{S} \cdot B = ie\Phi_S \tag{4.208}$$

Is the magnetic flux through the area swept by the particle.

However, it′ s heuristic to derive from Hamitonian:

$$U(t_f, t_i) = \int \mathrm{D}\boldsymbol{p}\mathrm{D}\boldsymbol{x} \exp[\frac{i}{\hbar} \int_{t_i}^{t_f} \mathrm{d}t(\boldsymbol{p} \cdot \dot{\boldsymbol{r}} - H(\boldsymbol{p}, \boldsymbol{r})] \tag{4.209}$$

The term in exponent involving $\boldsymbol{p}$ is:

$$-\frac{1}{2m\hbar}(\boldsymbol{p}^2 - 2\boldsymbol{p} \cdot (e\boldsymbol{A}(\boldsymbol{r}) + m\dot{\boldsymbol{r}} + e^2\boldsymbol{A}^2(\boldsymbol{r})) \tag{4.210}$$

To integrate over $\boldsymbol{p}$, we complete the square:

$$\boldsymbol{p}^2 - 2\boldsymbol{p} \cdot (e\boldsymbol{A} + m\dot{\boldsymbol{r}} = (p - (q\boldsymbol{A} + m\dot{\boldsymbol{r}}))^2 - (q\boldsymbol{A} + m\dot{\boldsymbol{r}})^2 \tag{4.211}$$

$$\frac{i}{2m\hbar}[(q\boldsymbol{A} + m\dot{\boldsymbol{r}})^2 - e^2\boldsymbol{A}^2] = \frac{i}{\hbar}\left[\frac{1}{2}m\dot{\boldsymbol{r}}^2 + q\dot{\boldsymbol{r}} \cdot \boldsymbol{A}(\boldsymbol{r})\right] \tag{4.212}$$

$$L = \frac{1}{2}m\dot{\boldsymbol{r}}^2 + q\dot{\boldsymbol{r}} \cdot \boldsymbol{A}(\boldsymbol{r}) \tag{4.213}$$

## 4.2 Gross-Pitaevskii Equation

For bosons with a contact interaction $\boldsymbol{V}(r, r') = g\delta(r - r')$:

$$\hat{H} = \int \mathrm{d}r \hat{\psi}^\dagger(r) \left[ -\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(r) \right] \hat{\psi}(r) + \frac{1}{2} \int \mathrm{d}r \, \mathrm{d}r' \hat{\psi}^\dagger(r) \hat{\psi}^\dagger(r') V(r, r') \hat{\psi}(r') \hat{\psi}(r) \tag{4.214}$$

$$\hat{H} = \int \mathrm{d}r \left[ \hat{\psi}^\dagger(r) \left[ -\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(r) \right] \hat{\psi}(r) + \frac{g}{2} \hat{\psi}^{\dagger(r)} \hat{\psi}^{\dagger(r)} \hat{\psi}(r) \hat{\psi}(r) \right] \tag{4.215}$$

$$S = \int \mathrm{d}t i\hbar \psi^* \partial_t \psi - \hat{H}(\psi^*, \psi) \tag{4.216}$$

$$\frac{\delta S}{\delta \psi^*} = i\hbar \partial_t \psi - \left( -\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(r) \right) \psi - g|\psi|^2 \psi = 0 \tag{4.217}$$

Rearranging gives the **GP Equation**:

$$i\hbar \frac{\partial \psi}{\partial t} = \left( -\frac{\hbar^2 \nabla^2}{2m} + V_{\text{ext}}(r) + g|\psi|^2 \right) \psi(\boldsymbol{r}, t) \tag{4.218}$$

## 4.3 Matsubara sums

Evaluate $I = 1/(\hbar\beta) \sum_{\omega_n} e^{i\omega\varepsilon} / (i\omega_n - x)$

Consider function $f(z) = \frac{e^{\varepsilon z}}{z - x} n_\eta(z)$ where $n_\eta(z) = \frac{1}{(e^{\hbar\beta z - \eta}}$.

The function $n_\eta(z)$ has poles at the Matsubara frequencies $z = i\omega_n$:

- Bosons ($\eta = 1$): $\omega_n = \frac{2n\pi}{\hbar\beta}$.
- Fermions ($\eta = -1$): $\omega_n = \frac{(2n+1)\pi}{\hbar\beta}$.

The residues of $n_\eta(z)$ at each $i\omega_n$ is $\frac{1}{\hbar\beta}$.

Integrate $f(z)$ over a large circle $C$ at infinity. For $0 < \varepsilon < \hbar\beta$, the factor $e^{\varepsilon z}$ and $n_\eta(z)$ ensure that the integral vanishes as the radius $R \to \infty$.

$$\oint_C \frac{\mathrm{d}z}{2\pi i} f(z) = \sum \mathrm{Res}(i\omega_n) + \mathrm{Res}(x) = 0 \tag{4.219}$$

- At $z = i\omega_n$, the residue is $\frac{e^{i\omega_n \varepsilon}}{i\omega_n x} \cdot \frac{1}{\hbar\beta}$.
- At $z = x$: The residue is $\frac{e^{\varepsilon x}}{e^{\hbar\beta x} - \eta}$.

$$\frac{1}{\hbar\beta}\sum_{\omega_n}\frac{e^{i\omega_n\varepsilon}}{i\omega_n - x} = -\frac{e^{\varepsilon x}}{e^{\hbar\beta x} - \eta} \tag{4.220}$$

## 4.4 Coherent-state path integral

Recall that for coherent states $|\psi\rangle$, the identity operator is expressed as:

$$\hat{1} = \int \frac{\mathrm{d}\psi^* \, \mathrm{d}\psi}{N} e^{-\psi^*\psi} |\psi\rangle\langle\psi| \tag{4.221}$$

When we time-slice the evolution operator:

$$\left\langle \psi_k \middle| e^{-\frac{i}{\hbar}\hat{H}(\hat{a}^\dagger, \hat{a})\varepsilon} \middle| \psi_{k-1} \right\rangle \approx \left\langle \psi_k \middle| \left(1 - \frac{i}{\hbar}\hat{H}(\hat{a}^\dagger, \hat{a})\varepsilon\right) \middle| \psi_{k-1} \right\rangle \tag{4.222}$$

$$\left\langle \psi_k \middle| \hat{H}(\hat{a}^\dagger, \hat{a}) \middle| \psi_{k-1} \right\rangle = \hat{H}(\psi_k^*, \psi_{k-1}) \langle \psi_k | \psi_{k-1} \rangle \tag{4.223}$$

Using the overlap formula for coherent states, $\langle \psi_k | \psi_{k-1} \rangle = e^{\psi_k^* \psi_{k-1}}$, we get:

$$\mathrm{LHS} \approx \exp\left(\psi_k^* \psi_{k-1} - \frac{i}{\hbar}\varepsilon H(\psi_k^*, \psi_{k-1})\right) \tag{4.224}$$

Generally, the action looks like this:

$$\frac{i}{\hbar} S[\psi, \psi^*](\{k\}) = \sum_{k=1}^{M} \left[\psi_k^* \psi_{k-1} - \psi_k^* \psi_k - \frac{i\varepsilon}{\hbar} H(\psi_k^*, \psi_{k-1})\right] \tag{4.225}$$

Where $-\psi_k^* \psi_k$ comes from the integral factor.

$$\psi_k^* \psi_{k-1} - \psi_k^* \psi_k = -\psi_k^* (\psi_k - \psi_{k-1}) \tag{4.226}$$

$$S[\psi, \psi^*] = \int_{t_i}^{t_f} \mathrm{d}t \left[i\hbar \psi^*(t) \dot{\psi}(t) - H(\psi^*(t), \psi(t))\right] \tag{4.227}$$

$$U\left(\psi_f^*, t_f; \psi_i, t_i\right) = e^{\psi_f^* \psi_f} \int \mathrm{D}\psi^* \, \mathrm{D}\psi \exp\left(\frac{i}{\hbar} S[\psi^*, \psi]\right) \tag{4.228}$$

## 4.5 Thermal traces and boundary conditions

$$\mathrm{tr}\left(\hat{O}\right) = \int \frac{\mathrm{d}\psi^* \, \mathrm{d}\psi}{N} e^{-\psi^*\psi} \left\langle \eta\psi \middle| \hat{O} \middle| \psi \right\rangle \tag{4.229}$$

Where $\eta = 1$ for bosons and $\eta = -1$ for fermions. Where the boundary condition can be evaluated as $\psi(\hbar\beta) = \eta\psi(0)$.

Based on previous deduction:

$$\left\langle \psi_k \middle| e^{-\frac{\varepsilon}{\hbar}\widehat{K}} \middle| \psi_{k-1} \right\rangle \approx e^{\psi_k^* \psi_{k-1}} e^{\left(-\frac{\varepsilon}{\hbar} K(\psi_k^*, \psi_{k-1})\right)} \tag{4.230}$$

$$\begin{aligned} Z = \lim_{M\to\infty} \int \prod_{k=1}^{M} \frac{\mathrm{d}\psi_k^* \mathrm{d}\psi_k}{N} \\ \exp\left( \sum_{k=1}^{M} \left[ \psi_k^* \psi_{k-1} - \psi_k^* \psi_k - \frac{\varepsilon}{\hbar} \hat{K}(\psi_k^*, \psi_{k-1}) \right] \right) \end{aligned} \tag{4.231}$$

$$S_E[\psi^*, \psi] = \int_0^{\beta\hbar} \mathrm{d}\tau \psi^*(\tau) \hbar \partial_\tau \psi(\tau) + H(\psi^*, \psi) - \mu \psi^*(\tau) \psi(\tau) \tag{4.232}$$

Which the $\mu\psi^*\psi$ is the mass term or chemical potential from $Z = \exp^{\left(-\beta\left(\hat{H} - \mu\hat{N}\right)\right)}$

For a non-interacting system, the Hamiltonian is quadratic: $\hat{K}_0 = \sum_\alpha (\varepsilon_\alpha - \mu) \hat{a}_\alpha^\dagger \hat{a}_\alpha$. THe functional integral decouples for each mode $\alpha$.

For a single mode alpha, the discrete action in is a quadratic form $\psi^\dagger S \psi$. Evaluate the matrix form:

- Diagonal: $\psi_k^* \psi_k$
- Off-diagonal: $\psi_k^* \psi_{k-1}$ and the $H$ term where $a = 1 - \frac{\varepsilon}{\hbar}(\varepsilon_\alpha - \mu)$.
- Corner: $\psi_0 = \eta\psi_M$, the first slice $\psi_1^* \psi_0$ actually becomes $\psi_1^*(\eta\psi_M)$.

$$\begin{aligned} Z_0 &= \lim_{M\to\infty} \prod_\alpha \int \prod_{k=1}^{M} \frac{\mathrm{d}\psi_k^{\alpha *} \mathrm{d}\psi_k^\alpha}{N} \\ &\quad \exp\left( -\sum_{k=1}^{M} \psi_k^{\alpha *} [(\psi_k^\alpha - \psi_{k-1}^\alpha)] + \varepsilon(\varepsilon_\alpha - \mu) \psi_{k-1}^\alpha \right) \\ &= \lim_{M\to\infty} \prod_\alpha \int \prod_{k=1}^{M} \frac{\mathrm{d}\psi_k^{\alpha *} \mathrm{d}\psi_k^\alpha}{N} \exp\left(-\psi^{\alpha\dagger} S^{(\alpha)} \psi^\alpha\right) \\ &= \lim_{M\to\infty} \prod_\alpha \left[\det(S)^{(\alpha)}\right]^{-\eta} \end{aligned} \tag{4.233}$$

Where $\varepsilon = \frac{\beta}{M}$.

$$a = 1 - \frac{\beta(\varepsilon_\alpha - \mu)}{M}, \quad S^{(\alpha)} = \begin{pmatrix} 1 & 0 & \cdots & -\eta a \\ -a & 1 & \cdots & 0 \\ 0 & -a & \ddots & 0 \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \quad \psi^\alpha = \begin{pmatrix} \psi_1^\alpha \\ \vdots \\ \psi_M^\alpha \end{pmatrix} \tag{4.234}$$

One can deduce the determinant by row reduction.

$$\det S^{(\alpha)} = 1 - \eta a^M = 1 - \eta \lim_{M\to\infty}\left(1 - \frac{\beta\Delta}{M}\right)^M = 1 - \eta e^{-\beta\Delta^\alpha}, \quad \Delta^\alpha = \varepsilon_\alpha - \mu \tag{4.235}$$

$$Z_0 = \prod_\alpha (1 - \eta \exp(-\beta(\varepsilon_\alpha - \mu))^{-\eta} \tag{4.236}$$

From previous derivation of each single particle mode $\alpha$:

$$Z_{0,\alpha} = \left[\det S^{(\alpha)}\right]^{-\eta} \tag{4.237}$$

$$Z_0 = \exp\left(\sum_\alpha \ln\left[\det S^{(\alpha)}\right]^{-\eta}\right) = \exp\left(-\eta \sum_\alpha \ln\det S^{(\alpha)}\right) \tag{4.238}$$

We now use the fundamental identity for any square matrix $M$:

$$\det M = \exp(\operatorname{tr}\ln M) \tag{4.239}$$

$$Z_0 = \exp\left(-\eta \sum_\alpha \operatorname{tr}\ln S^{(\alpha)}\right) \tag{4.240}$$

We apply Lagrange transformation to the action.

$$Z[J^*, J] = \int \mathrm{D}\psi^*\,\mathrm{D}\psi \exp\left(-\frac{S[\psi^*, \psi]}{\hbar} + \sum_i (J_i^*\psi_i + \psi_i^* J_i)\right) \tag{4.241}$$

In the discrete case:

$$-\sum_{jk} \psi_j^* S_{jk} \psi_k + \sum_i (J_i^*\psi_i + \psi_i^* J_i) \tag{4.242}$$

This is the trick makes the Gaussian integral solvable. We shift the variables of $\psi$ and $\psi^*$ to elminate the linear source terms.

$$\phi = \psi - S^{-1}J, \quad \phi^* = \psi^* - J^* S^{-1} \tag{4.243}$$

$$\psi^* S\psi - J^*\psi - \psi^* J = \phi^* S\phi + J^* S^{-1} J \tag{4.244}$$

$$Z[J^*, J] = Z_0 \exp(J^* S^{-1} J) \tag{4.245}$$

Note that $Z_0 = [\det S]^{-\eta}$ as derivd previously.

$$G_{qr} = -\langle \psi_q \psi_r^* \rangle = -\frac{1}{Z_0} \int \mathrm{D}\psi^* \mathrm{D}\psi \psi_q \psi_r^* \exp\left(-\frac{1}{\hbar} S_E\right) \tag{4.246}$$

Using the property of exponentials, we can replace the variables $\psi$ and $\psi^*$ with the deriviatives with respect to their sources:

$$\psi_q \to \frac{\partial}{\partial J_q^*}, \quad \psi_r^* \to \frac{\partial}{\partial J_r} \tag{4.247}$$

$$G_{qr} = -\frac{\eta}{Z_0} \frac{\partial^2}{\partial J_q^* \partial J_r} \left[ Z_0 \exp\left( \sum_{jk} J_j^* \left(S^{-1}\right)_{jk} J_k \right) \right]_{J=J^*=0} = -\left(S^{-1}\right)_{qr} \tag{4.248}$$

When we differentiate $J_j^* \left(S^{-1}\right)_{jk} J_k$ with respect to $J_q^*$, $J_r$, only the term where $j = q$ and $k = r$ survivies. Then we requires to find the elements of $S^{-1}$ that $S \cdot G = -I$.

$$\sum_j S_{kj} G_{jr} = -\delta_{kr} \tag{4.249}$$

$$G_{1r} - \eta a G_{Mr} = -\delta_{1r}, \quad k = 1 \tag{4.250}$$

$$G_{kr} - a G_{k-1} r = -\delta_{kr}, \quad k > 1 \tag{4.251}$$

Away from the source at $k = r$, the equation is simply $G_{kr} = a G_{k-1r}$.

In general for $k \geq r$:

$$G_{r+1,r} = a G_{rr} \Rightarrow G_{r+2,r} = a^2 G_{rr} \Rightarrow G_{kr} = a^{k-r} G_{rr} \tag{4.252}$$

For $k < r$:

$$G_{1r} = \eta a G_{Mr} = \eta a \left(a^{M-r} G_{rr}\right) = \eta a^{M-r+1} G_{rr} \tag{4.253}$$

$$G_{kr} = a^{k-1} G_{1r} = a^{k-1} \left(\eta a^{M-r+1} G_{rr}\right) = \eta a^{M-r+k} G_{rr} \tag{4.254}$$

To anchor these formulas, we evaluate diagonal term:

$$\begin{aligned} G_{rr} - a G_{r-1r} &= -1 \\ G_{rr} - a\left(\eta a^{M-r+(r-1)} G_{rr}\right) &= -1 \\ G_{rr} - \eta a^M G_{rr} &= -1 \\ G_{rr}\left(1 - \eta a^M\right) &= -1 \\ G_{rr} &= -\frac{1}{1 - \eta a^M} \end{aligned} \tag{4.255}$$

When $M \to \infty$, $a^M \to \exp(-\beta\Delta)$, and $a^{k-r} \to \exp\left(-\left(\tau_q - \tau_r\right)\Delta/\hbar\right)$.

Recall the density distribution:

$$n_\alpha = \frac{1}{e^{\beta\Delta} - \eta}, \quad 1 + \eta n_\alpha = \frac{1}{1 - \eta e^{-\beta\Delta}} = G_{rr} \tag{4.256}$$

For $\tau_q > \tau_r$ ($k > r$):

$$G_{qr} = a^{k-r} G_{rr} = e^{-\Delta\left(\tau_q - \tau_r\right)/\hbar}(1 + \eta n_\alpha) \tag{4.257}$$

For $\tau_q < \tau_r$ ($k < r$):

$$\begin{aligned} G_{qr} = \eta a^{M-(r-k)} G_{rr} &= \eta e^{-\beta\Delta} e^{-\Delta\left(\tau_q - \tau_r\right)/\hbar}(1 + \eta n_\alpha) \\ &= -e^{-\Delta\left(\tau_q - \tau_r\right)/\hbar}(\eta n_\alpha) \end{aligned} \tag{4.258}$$

$$G_0\left(\tau_q, \tau_r\right) = -e^{-\Delta\left(\tau_q - \tau_r\right)/\hbar}\left[\theta\left(\tau_q - \tau_r\right)(1 + \eta n_\alpha) + \eta\theta\left(\tau_r - \tau_q\right)n_\alpha\right] \tag{4.259}$$

We already now that $G_0^{(\alpha)} = -\left[S^{(\alpha)}\right]^{-1}$ that:

$$Z_0 = \exp\left(-\eta \sum_\alpha \operatorname{tr}\ln\left(-\left[G_0^{(\alpha)}\right]^{-1}\right)\right) \tag{4.260}$$

I we look at its structure in discrete time-slice representation, it looks exactly like the matrix $S$ but the scalar $a$ replaced by the operator $\hat{a} = 1 - \beta\left(\hat{h}_0 - \mu\right)/M$, where $\hat{h}_0$ is the single particle Hamiltonian.

Since the grand potential is $\Omega = -k_B T \ln Z$, the formula gives:

$$\Omega_0 = \eta k_B T \operatorname{tr}\ln\left(-\hat{G}_0^{-1}\right) \tag{4.261}$$

# 5 Perturbative expansion

Suppose $S = S_0 + S_V$.

$$Z = \int \mathrm{D}\psi^* \, \mathrm{D}\psi \exp\left(-\frac{S_0 + S_V}{\hbar}\right) \tag{5.262}$$

$$Z = Z_0 \cdot \frac{Z}{Z_0} \tag{5.263}$$

By definition, the ratio on the right is the expectation alue of the operator $e^{-S_V/\hbar}$ with respect to the non-interacting weight $e^{-S_0/\hbar}$.

$$Z = Z_0 \left\langle \exp^{-S_V/\hbar} \right\rangle_0 \tag{5.264}$$

However, it' s not applicable to directly solve it, we expand the exponential function:

$$\langle \exp(-S_V/\hbar) \rangle = \left\langle \sum_{n=0}^{\infty} \frac{1}{n!} \left( -\frac{S_V}{\hbar} \right)^n \right\rangle_0 \tag{5.265}$$

$$\frac{Z}{Z_0} = \sum_{n=0}^{\infty} \left( -\frac{1}{\hbar} \right)^n \frac{1}{n!} \langle S_V^n \rangle_0 \tag{5.266}$$

It' s directly that for a set of complex variables:

$$\frac{\int \mathrm{dz}^* \, \mathrm{dz} \left( z_i z_j^* \right) e^{-\mathrm{z}^* S \mathrm{z}}}{\int \mathrm{dz}* \, \mathrm{dz} e^{-\mathrm{z}* S \mathrm{z}}} = \left( S^{-1} \right)_{ij} \tag{5.267}$$

$$Z[J^*, J] = \int \mathrm{D}\psi^* \mathrm{D}\psi \exp \left( -\frac{S[\psi^*, \psi]}{\hbar} + \sum_i (J_i^* \psi_i + \psi_i^* J_i) \right) = Z_0 \left\langle J^* S^{-1} J \right\rangle \tag{5.268}$$

$$\left\langle \psi_i \psi_j^* \right\rangle = \frac{1}{Z_0} \frac{\partial^2 Z}{\partial J_i^* \partial J_j} \Bigg|_{J=0} = \frac{\partial^2}{\partial J_i^* \partial J_i} \exp \left( \sum_{ab} J_a^* \left( S^{-1} \right)_{ab} J_b \right) \tag{5.269}$$

$$\left\langle \prod_{i=1}^{n} \psi_i \prod_{j=n}^{1} \psi_j^* \right\rangle_0 = \sum_{\sigma} \eta^{|\sigma|} \prod_{i=1}^{n} (-G_0(i, \sigma(i)) \tag{5.270}$$

For connection of element $i$ and $\sigma(i)$. Which is the product generated by all permutation $\sigma$.

- Contraction, i.e. a Green' s function for non-interactive system, is repsented by a propagator line: 
- Interaction element, is represented by a vertex, usually, the four body is two incoming and two outgoing lines: